

Amit Singer's

Mathematics for Cryo-Electron Microscopy



Figure 1: **Amit Singer, Author of the Paper**

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1 Introduction

In the last ten years or so, single-particle cryo-electron microscopy, or cryo-EM, has completely changed the field of structural biology. Cryo-EM involves rapidly chilling diluted biomolecules in a thin layer of amorphous ice, after which, it uses an electron microscope to scan them in various orientations. In essence, a two-dimensional "particle image" is a tomographic projection of the 3D electrostatic potential of the molecule integrated along the direction of the beam. In contrast to the X-ray crystallography or NMR, cryo-EM can determine the structures of individual particles close to physiological circumstances without any need of crystals.

A "resolution revolution" has sprung from this, as numerous complexes that were previously not solvable by crystallography are now very easily produced at resolutions better than 4 Å. Notably, cryo-EM won the 2017 Nobel Prize in Chemistry for its discovery, and was chosen as *the Method of the Year* by the *Nature Methods* in 2015.

However, cryo-EM presents very daunting data-analysis challenges. The raw projection images have very very low signal-to-noise ratio (SNR) because only a very small electron dose can be used to prevent damage due to the radiation involved. Crucially, the 3D orientation of each particle (a rotation $R_i \in SO(3)$) is unknown.

Mathematically, Singer models each measured image I_i as a convolution of the projected density with a point spread function plus noise. In a simplified form, one can write:

$$I_i(\mathbf{x}) = (h * (P R_i \phi))(\mathbf{x}) + \text{noise},$$

where:

- $\phi(x)$ is the unknown 3D potential,
- R_i is the unknown rotation,
- P projects along the z -axis,
- h is the microscope's point-spread function.

Recovering ϕ from numerous noisy photos is a **nonlinear inverse problem** as the orientations R_i are not provided. As a result of this, any effective reconstruction technique needs to simultaneously infer the orientations of every particle and the molecular structure.

Mathematics is quite essential in this situation. Singer highlights how cryo-EM combines tools from various areas like **tomography**, **signal processing**, **statistics**, and **geometry**. His study aims to present these concepts to mathematicians. Throughout the book, Singer describes how the practical challenge of creating a 3D model from millions of noisy 2D images is made possible by abstract notions, for example: group invariants, spherical harmonics, and Fourier transforms.

In order to establish the scene and cruise ahead, we first describe Singer's background. Next, we highlight the main points in the paper. Lastly, I offer my own thoughts on the concepts I discovered and the elements that most inspired me.

2 About the Scientist

Amit Singer is a well-known mathematician who, particularly in structural biology, connects theory with practice. In addition to serving on the executive committee of Princeton University's Program in Applied and Computational Mathematics (PACM), he is the *Herbert E. Jones, Jr. '43 University Professor of Mathematics*. He was a Gibbs Assistant Professor at Yale University, New Haven, US, from 2005 to 2008 before joining the Princeton faculty in 2008, originally as an assistant professor.

Singer graduated from Tel Aviv University, in Israel, with a B.Sc. in mathematics and physics in 1997 and a Ph.D. in applied mathematics in 2005. Israel awarded him the Nessyahu Prize for Best Mathematics Dissertation in 2005 for his Ph.D. thesis on diffusion theory of ion channels. He also completed his graduate studies at Rush University Medical Center early in his career.

Singer's research spans theoretical and computational aspects of *data science* and *structural biology*. He has made fundamental contributions to fields such as:

- Dimensionality reduction (diffusion maps)
- Spectral graph theory
- Sensor networks
- Cryo-EM imaging

In cryo-EM specifically, Singer has developed novel algorithms that combine representation theory and synchronization to recover 3D structure from noisy 2D images. He leads the ASPIRE software project (Algorithms for Single Particle Reconstruction), which automates the cryo-EM data analysis with minimal amount of human intervention.

Throughout his career, Singer has received many honors:

- *Simons Investigator* in Applied Mathematics (2012)
- *Moore Foundation Investigator in Data-Driven Discovery* (2014)
- NSF CAREER Award/PECASE (2010)
- Alfred P. Sloan Research Fellowship (2010)
- Israel's Nessyahu Prize (2007)

He has co-authored over 100 scientific papers and served on editorial boards of leading journals, including the SIAM Journal on Imaging Sciences and Applied and Computational Harmonic Analysis. In addition to PACM, Singer sits on the executive committee of Princeton's Center for Statistics and Machine Learning (CSML), reflecting his leadership at the interface of mathematics, computation, and biology.

In summary, Prof. Singer is an accomplished applied mathematician whose:

- *Academic background* (Tel Aviv B.Sc./Ph.D.)
- *Research* (algorithms for high-dimensional noisy data, cryo-EM structure determination, manifold learning)
- *Leadership roles* (PACM, CSML, editorial boards)

These establish him as an authority on the mathematics of cryo-EM.

3 What Was Done in the Paper

Singer’s paper is a concise **review** (ICM 2018 invited lecture) of the mathematical foundations of single-particle cryo-EM. Its goal is *not* to introduce new algorithms, but to explain the key ideas and limitations behind existing methods. The paper progresses through the imaging model, classical reconstruction strategies, challenges from noise, and advanced statistical approaches. The main points are as follows:

- **Imaging model and inverse problem:** Singer begins by formulating the forward model. Let $\phi(\mathbf{r})$ be the unknown 3D electrostatic potential of the molecule. Each image $I_i(x, y)$ arises by rotating the molecule by some unknown $R_i \in SO(3)$, projecting it along the z -axis, convolving with the microscope’s point-spread function, and adding noise. Symbolically one can write

$$I_i = h * (PR_i\phi) + \text{noise},$$

where P integrates ϕ along z , h is the PSF, and the R_i are unknown. Thus the *basic cryo-EM inverse problem* is: given many noisy images $\{I_i\}$, estimate ϕ without knowing $\{R_i\}$. This unknown-orientation makes the problem **nonlinear**. Singer also notes that when molecules exist in multiple conformations (heterogeneity), the inverse problem becomes even more challenging, though his paper focuses on the simpler single-structure setting.

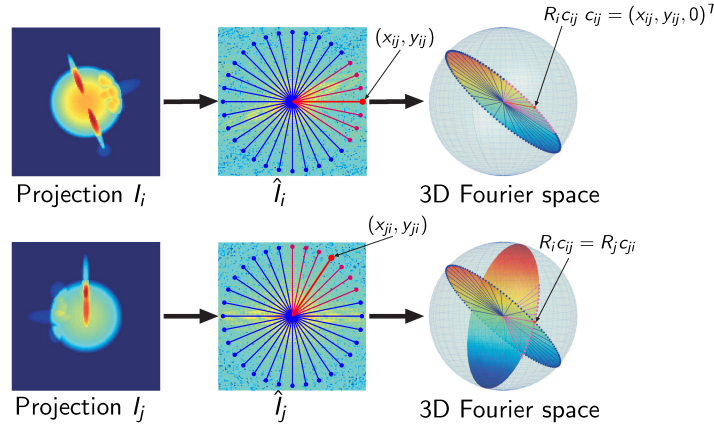


Figure 2: Fourier slice theorem and common line property.

- **Fourier projection-slice theorem:** A cornerstone is the projection-slice theorem: the 2D Fourier transform of a projection equals the restriction of the 3D Fourier transform $\mathcal{F}_3[\phi]$ to a central plane perpendicular to the viewing direction. In formulas, if $\hat{l}_i(\omega_x, \omega_y)$ is the 2D Fourier transform of I_i , then

$$\hat{l}_i(\omega_x, \omega_y) = (\mathcal{F}_3[\phi])(R_i^T(\omega_x, \omega_y, 0))$$

for each frequency (ω_x, ω_y) . Geometrically, each image provides a planar “slice” through the 3D Fourier space of the molecule.

- **Common lines and angular reconstitution:** Singer explains the *common-line* property: any two distinct central Fourier planes intersect in a line through the origin. Hence any pair of projection images must share a common line in their Fourier transforms where the values agree. Practically, one can scan all candidate lines in image i against lines in image j and pick the pair with maximum cross-correlation. Each identified common line fixes two of the three Euler angles relating images i and j . To get the third angle, one introduces a third image: with three images, all pairwise common lines form three great circles that geometrically determine the relative rotations (up to a global mirror). This classic procedure is known as **angular reconstitution** (Vainshtein–Goncharov 1986, van Heel 1987). Singer illustrates these ideas in *Figure 3* of the paper. He also points out a subtlety: a molecule and its mirror image (enantiomer) produce identical projection sets, so single-particle cryo-EM inherently cannot distinguish handedness.

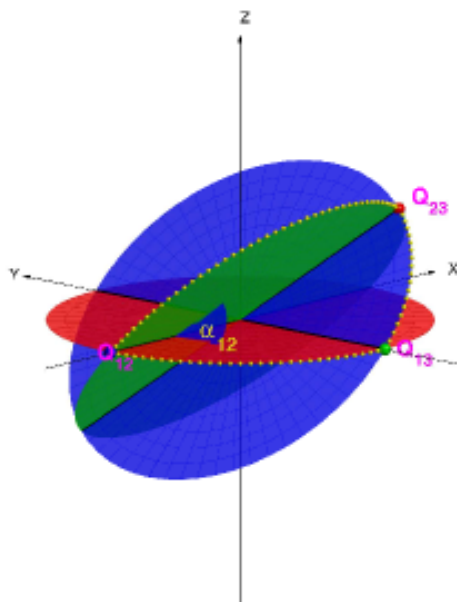


Figure 3: Angular reconstitution.

- **Clean vs. noisy data:** For ideal, noise-free images, the common-line method can correctly assign orientations. However, Singer emphasizes that **experimental images are very noisy**. He presents simulations showing that the probability of correctly identifying a common line plummets as SNR drops. For typical cryo-EM SNR (~ -3 to -6 dB), only a small fraction of common lines are detected correctly. Since angular reconstitution requires three correct line matches, its success rate is extremely low in practice. Singer thus notes that sequential alignment by pairwise common lines accumulates errors quickly, making direct approaches unreliable at realistic noise levels.
- **Denoising via 2D classification:** To cope with noise, practitioners often group images by similarity (2D classification), align within each class, and average to boost SNR. Singer

mentions this strategy: by averaging dozens of supposedly similar views, one can create cleaner “class averages” which then yield more robust common lines. This helps for large particles at moderate resolution, but Singer warns that for small molecules or extremely low SNR it still fails.

- **Iterative refinement (MLE/EM):** In Section E, Singer reviews the dominant modern approach: iterative refinement with an expectation-maximization (EM) framework. One starts with an initial 3D model and alternates between (1) projecting the model to generate template images, (2) comparing each noisy image to all templates to assign (soft) rotations, and (3) reconstructing a new 3D model from these assignments. This process (embodied in software like RELION) essentially performs maximum-likelihood estimation of the volume while treating the orientations as hidden variables. Singer notes its practical success but also its drawbacks: it can be trapped in local optima, requires a good starting model to avoid bias, and is computationally expensive. He raises the important question of sample complexity: how many images are needed to reconstruct at a given resolution, noting that traditional MLE theory doesn’t directly apply here because the number of parameters grows with image count.
- **Kam’s autocorrelation method (moments):** A highlight of the paper is the discussion of Kam’s 1980s method. Kam proposed bypassing orientation estimation entirely by computing the covariance (and higher-order correlations) of all 2D images. Singer explains that under the assumption of uniformly random orientations, the covariance of the images is directly related to the autocorrelation of the unknown 3D structure. In essence, one uses steerable PCA on the image Fourier transform to extract matrices C_ℓ that factor as $C_\ell = F_\ell F_\ell^*$ (where F_ℓ encodes the radial functions of ϕ in a spherical harmonic expansion). The factor F_ℓ can be found up to an unknown orthogonal matrix. By collecting enough information (e.g. via triple correlations), these unknown factors can be determined and the structure recovered. Singer points out Kam’s key advantages: it requires only **one pass through the data**, is extremely fast, and has no need for an initial model. He also notes limitations: it assumes uniform orientation distribution (often violated), perfect centering of particles, and a very large number of images to estimate high-order statistics accurately. Singer’s recent work (cited in the paper) has improved the covariance estimation using techniques like steerable PCA and optimal shrinkage, but he acknowledges these methods remain sensitive to practical issues.
- **Multi-Reference Alignment (toy model):** To illustrate fundamental statistical limits, Singer describes a simpler 1D toy problem called multi-reference alignment (MRA). Here one observes a 1D signal x that is randomly shifted (cyclically) and corrupted by noise: $y_i(k) = x(k + g_i) + \varepsilon_i(k)$. The task is to recover x without knowing the random shifts g_i . Singer shows that this model captures the essence of cryo-EM: at high SNR one can estimate shifts (orientations) and align normally; but at very low SNR it becomes impossible to identify the shifts correctly. Crucially, one can still recover x by estimating *invariant* features: the average, the power spectrum, and the bispectrum. These moments are shift-invariant summaries of the data. For example, the bispectrum (a triple product in Fourier space) retains the missing phase information needed for unique recovery. Singer

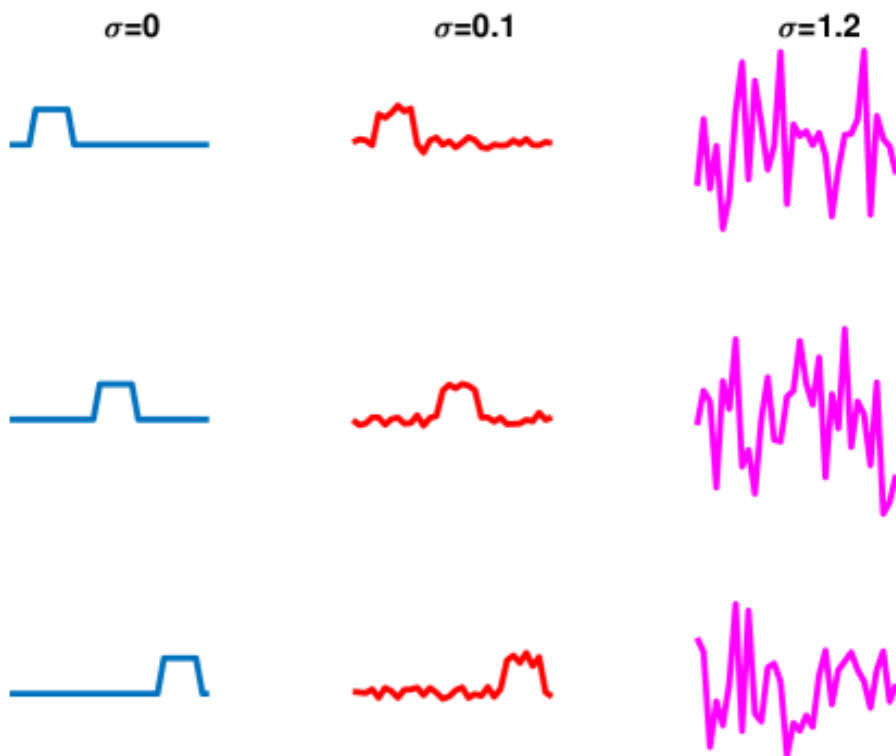


Figure 4: Multi-reference alignment of 1D periodic signals under varying noise levels (σ).

explains that with enough observations, one can solve for x from these invariants even when individual shifts are unrecoverable. This analogy clarifies why in cryo-EM one might abandon explicit orientation assignment and instead use covariance (2nd moment) and higher moments as in Kam’s method.

- **Summary of open challenges:** In his final section, Singer emphasizes that many computational problems in cryo-EM remain **open**. These include handling extreme noise for small particles, correcting for beam-induced motion, dealing with continuous heterogeneity (flexible molecules), and integrating prior information. He stresses that cryo-EM problems draw upon a remarkable variety of mathematical fields: in his words, “*representation theory, tomography and integral geometry, high-dimensional statistics, random matrix theory, information theory, algebraic geometry, signal processing, manifold learning, numerical linear algebra, and fast algorithms*” all play a role. This review therefore does not propose new algorithms, but surveys the existing ones and highlights where mathematics has made an impact or where more work is needed.

In summary, Singer’s paper carefully lays out the **mathematical tools and challenges** in single-particle cryo-EM. Key concepts include the Fourier projection-slice theorem and common-line method for orientation assignment, the limitations imposed by high noise, the maximum-likelihood EM algorithms for refinement, and the method-of-moments (Kam) approach that bypasses orientations. It serves as an excellent guide to *what is being done* to reconstruct 3D structures from cryo-EM data and *why* it works (or sometimes doesn’t).

4 What I Learned – In My Own Words

I gained a deeper grasp of how advanced mathematics resolves the cryo-EM reconstruction problem after reading Singer's work. I specifically learnt four quite important lessons:

- **Fourier slice theorem in practice.** Through the 3D transform's inception, Singer concretizes the notion that the Fourier transform of every 2D image resides on a plane. In frequency space, pairs of pictures connect along "common lines," and unknown rotations R_i become unknown planes. It feels sleek and very natural to match those 1D lines instead of the full, large images.
- **Noise fundamentally limits alignment.** According to his simulations, the majority of common lines are misdetected at realistic SNR (around -3 dB). In order to increase SNR prior to the alignment, practitioners cluster and average images since the sequential angular reconstruction fails in that case.
- **Refinement iteratively as EM.** Contemporary processes (like RELION) alternate between projecting a temporary 3D model, assigning soft rotations by matching those projections to noisy pictures, and then reconstructing an updated model. This maximum-likelihood (EM) method can become stuck in local optima and needs a strong starting model to prevent bias.
- **Moments-of-method (Kam's strategy)** One calculates the global statistics of all images, such as covariance and higher moments, rather than estimating orientations. The covariance encodes the 3D density's autocorrelations under uniform orientation assumptions, which can be recovered using steerable PCA up to orthogonal factors. The remaining uncertainties are then resolved using triple correlations (bispectrum). This quick, one-pass method does not require an initial model, but it does require a large number of photos and precise centering.

In addition, the paper's 1D toy model (multi-reference alignment) explained why invariant properties, such as mean, power spectrum, and bispectrum, are adequate for signal recovery in the event that direct alignment is unsuccessful. Singer's use of *spherical harmonics* in Kam's theory appeared abstract at first, but his explanation revealed that a matrix equation $C_\ell = F_\ell F_\ell^*$ with factors known up to unknown rotations is produced for each angular frequency ℓ . Those rotations are then resolved by higher moments.

Cryo-EM is a highly interdisciplinary problem that integrates group theory (rotations), harmonic analysis, statistics (covariance, EM), optimization (alignment), and even algebraic geometry, according to Singer's review. My mathematical toolset has expanded thanks to his lucid analogies, which have helped me understand why certain approaches are effective, where they fall short, and how alignment-based tactics might be enhanced by global statistical approaches.

5 What Amazed Me – The Power of Mathematics in Biology

Several aspects of Singer’s paper inspired me deeply:

- **Mathematics meets reality:** The direct connection between *abstract* mathematics and *real-world* biology astounds me. Singer specifically cites algebraic geometry and representation theory as essential domains for cryo-EM. It is incredibly motivating to know that ideas like **group representations** and **manifold learning** are at the heart of Nobel-winning research. The fact that something I learned in a pure math class—like projecting on spherical harmonics—is essential to visualizing proteins leaves me spell bound
- **Common line elegance:** The common-line concept is lovely in and of itself. In Fourier space, a line will always cross two slanted replicas of a three-dimensional object. This geometric image caught my attention since it simplifies a hidden 3D orientation issue into 2D matching lines, which is much easier to see. The fact that the complete 3D rotating frame is locked down by just three images is elegant. I found this angular reconstitution’s simplicity to be academically astounding.
- **Moments over alignment:** Kam’s method—that you don’t have to align the particles to perceive the structure—may be the most unexpected. According to Singer, recovering the molecule (up to a few orthogonal components) only requires calculating the covariance of all images. I was astounded by this. It indicates that the 3D structure is seen simply by "averaging" in a high-dimensional manner. It is remarkable to think that all of the missing orientation information is encoded by high-order statistics (triple correlations, bispectrum). It exhibits a certain mathematical elegance in that the only dependable signal in the presence of severe noise is found in aggregated invariants. This was made evident by the paper’s comparison to multi-reference alignment: the correlations encode the entire puzzle even when we are unable to determine where each component goes.
- **Handedness ambiguity:** The subtle fact that cryo-EM cannot differentiate a molecule from its mirror image also caught me off guard. Singer notes that the reconstruction is essentially ambiguous up to a global reflection since reflections of the genuine structure yield identical projection data. This is a profound illustration of mathematical symmetry that surprised me a lot. It reminds me of the interesting fact that this approach cannot detect the inbuilt-chirality of nature.
- **In real life impact:** As I read Singer, I reminded myself that these mathematical concepts had already produced real advances. According to the article, cryo-EM has solved structures such as the TRPV1 ion channel (Nature 2013) and many complexes with a resolution of 3–4 Å. The fact that both theory and algorithmic creativity produced these outcomes stupefies me. The ASPIRE software was created by Singer’s own team, and these techniques probably helped solve a lot of proteins. It gives the task a very realistic feel to know that mathematicians like Singer contribute to these kinds of outcomes.
- **Multidisciplinary approach:** Lastly, Singer’s writing and position highlight interdisci-

plinary cooperation. The field of cryo-EM encompasses biology, physics, computer science, and mathematics. This attitude of interdisciplinary collaboration inspires me. For instance, Singer talks about collaborating with biologists to apply algorithms to actual data and with physicists to comprehend the optics of the microscope. It is incredibly inspiring to observe how these fields converge in real-world applications since it demonstrates how theory may lead to transformative discoveries.

In conclusion, I learned about the close connection that exists among abstract mathematics and real-world biological imaging from Amit Singer’s “*Mathematics for Cryo-EM*”. It provided elegant solutions (common lines, method of moments) and clarified complex ideas (such as rotational alignment and statistical reconstruction). The elegance of the underlying mathematics inspires me, and I am motivated by the practical achievements they make possible. The paper’s blend of geometry, statistics, and computation serves as the ideal example of how complex mathematical concepts could illuminate the molecular structure of life.

6 References

Amit Singer, “*Mathematics for Cryo-Electron Microscopy*”

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